

ACADEMIC PROJECT

Simple Calculator Application

Android Studio project using Java and XML

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Android Studio Project

ACADEMIC
PROJECT

Simple Calculator App

A small Android project built with Java and XML.

PLATFORM
Android

DEVELOPMENT
Android Studio

LANGUAGE
Java

Academic Project Presentation

Prepared from the project code and actual app output.

Project overview

What I built

I made a simple Android calculator that takes input from the keypad and calculates mathematical expressions on the same screen.

The main idea

- Keep the interface simple and easy to use.
- Support common operations, decimals and brackets.
- Show the result while entering the expression.

Project snapshot

Java handles the button clicks and calculation logic.

XML defines the calculator screen and button layout.

The app runs on an Android emulator / device.

Why I made it

01 Practice Android UI

Design a working screen with XML and Material Buttons.

02 Practice Java events

Connect each button with one click handler.

03 Practice problem solving

Turn a typed expression into a correct numerical result.

04 Build a useful app

Create something small that can actually be used for everyday calculations.

Technologies used

ANDROID STUDIO

Used to create, run and test the app.

JAVA

Used for button handling and calculator logic.

XML

Used to design the calculator interface.

MATERIALBUTTON

Used for the rounded calculator buttons.

TEXTVIEW

Used to show the expression and result.

Interface design

How the screen is arranged

- Two TextViews at the top show the expression and final result.
- The keypad is placed at the bottom in five rows.
- Number and control buttons use a rounded Material design.
- Operator and equals buttons are visually highlighted.

5 ROW
KEYPAD

2 DISPLAYS

Main features

Basic maths

Addition, subtraction, multiplication and division.

Brackets

Expressions can include () for grouped calculations.

Decimals

Decimal values can be entered using the dot button.

C button

Removes the last character from the current expression.

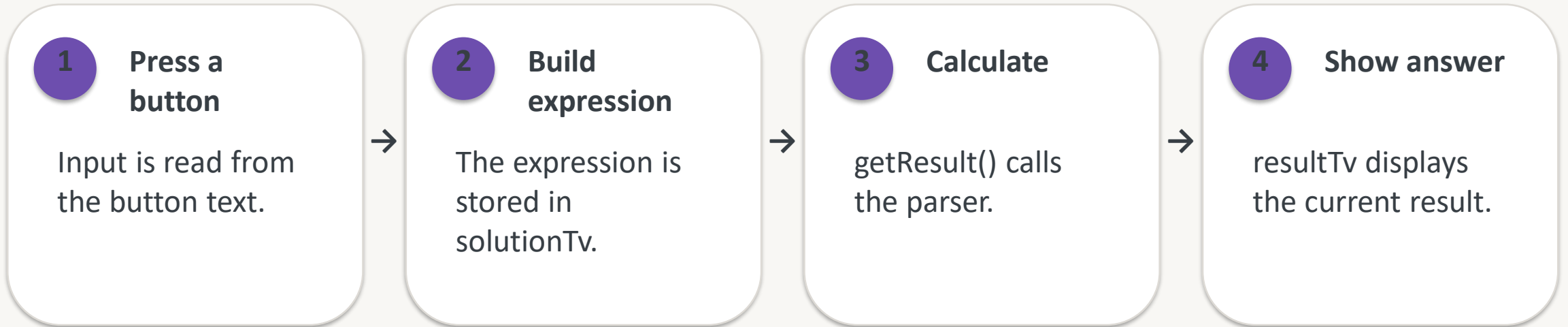
AC button

Clears the full expression and resets the result to 0.

Live result

The app updates the result as a valid expression is entered.

How the calculator works



Example **5 + 5** **=** **10**

The screenshot shows the same test running in the Android emulator.

Java code structure

MainActivity connects the UI with the logic.

```
public class MainActivity extends AppCompatActivity
    implements View.OnClickListener {

    private TextView resultTv;
    private TextView solutionTv;

    private void setClickListener(MaterialButton...
    buttons) {
        for (MaterialButton button : buttons) {
            button.setOnClickListener(this);
        }
    }
}
```

What this part does

- Finds the TextViews and all calculator buttons with findViewById().
- Uses one click listener for the buttons instead of writing separate listeners.
- Sends the typed expression to the calculation method.

This keeps the code easier to follow.

Calculation logic

The parser works in three levels

parseExpression() handles + and -

parseTerm() handles * and /

parseFactor() handles numbers, decimals, brackets and signs

Example: $10 + 6 * 2$

1. $6 * 2$ is calculated first.
2. Then $10 + 12$ is calculated.

Result = 22

The code also checks for division by zero and returns "Err" when an invalid calculation occurs.

Testing and error handling

TEST	OUTPUT	PURPOSE
5 + 5	10	Normal calculation
2.5 * 4	10	Decimal input
(8 + 2) / 2	5	Brackets
10 / 0	Err	Division by zero
AC	0	Clear all input
C	Removes last character	Delete one input

The app handles errors inside getResult() and keeps the interface running.

Actual output



The app is running successfully.

- The expression $5 + 5$ is entered through the keypad.
- The result 10 appears on the display.
- The emulator view also shows the final calculator interface.

**OUTPUT
VERIFIED**

Screenshot taken from the Android Studio emulator.

What I learned

Through this project, I practiced:

- Designing an Android interface with XML.
- Handling user input with Java and click events.
- Writing a simple expression parser and checking errors.
- Running and testing the app on an Android emulator.

Next improvements

- Add calculation history.
- Add dark mode and themes.
- Add scientific calculator functions.
- Improve input validation for longer expressions.

THANK YOU

Simple, practical, and focused on the core Android development concepts I wanted to practice.